How-To Library

Informatica

Deploy Axon Data Governance 7.2.3 on the AWS U.S. Intelligence Community Marketplace

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Abstract

You can deploy Axon Data Governance on the Amazon Web Services (AWS) U.S. Intelligence Community Marketplace. This deployment reference includes step-by-step instructions for deploying Axon Data Governance on the Amazon Web Services (AWS) U.S. Intelligence Community Marketplace. It also includes information on prerequisites and troubleshooting.

Supported Versions

- Axon Data Governance 7.2.3

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Apache Log4j RCE Vulnerability for Axon Data Governance

The Log4j vulnerability is addressed in the following ways:

- The Elasticsearch service in Axon is updated to version 6.8.22, which uses Log4j version 2.17.0.

- The Axon Agent uses Log4j version 2.17.1.

Overview

Automated reference deployments use AWS CloudFormation templates to launch, configure, and run the AWS compute, network, storage, and other services required to deploy a specific workload on AWS.

Intended Audience

This document is for users who are responsible for deploying Axon Data Governance on the AWS U.S. Intelligence Community Marketplace.

As a user with administrator privileges to deploy applications on AWS U.S. Intelligence Community Marketplace, we assume that you are familiar with the following AWS resources:

- Amazon VPC
- Amazon EC2
- Amazon S3
- Elastic IP addresses
- Cloud formation
- RDS
- Internet gateway
- NAT gateway
- Route table
- Security group

You must be familiar with Axon Data Governance, and we recommend that you are familiar with the concepts of IP CIDR and public and private IP addresses.

Knowledge Resources

Use the following resources to become familiar with the relevant AWS concepts and Axon Data Governance:

AWS Services

If you are new to AWS, see Getting Started with AWS.

For AWS Services documentation, refer to the following:

- AWS CloudFormation
- Amazon EBS
- Amazon EC2
- Amazon VPC
- Amazon S3
- Elastic IP Addresses

Reference Deployment

Refer to the AWS deployment at AWS Deployment home page.

Axon Data Governance Documentation

For Axon Data Governance 7.2.3 documentation, see the Informatica Documentation Portal.

Costs and Licenses

The AWS CloudFormation template for this deployment includes configuration parameters that you can customize. Some of these settings, such as instance type, will affect the cost of deployment. For cost estimates, see the pricing pages for each AWS service you plan to use.

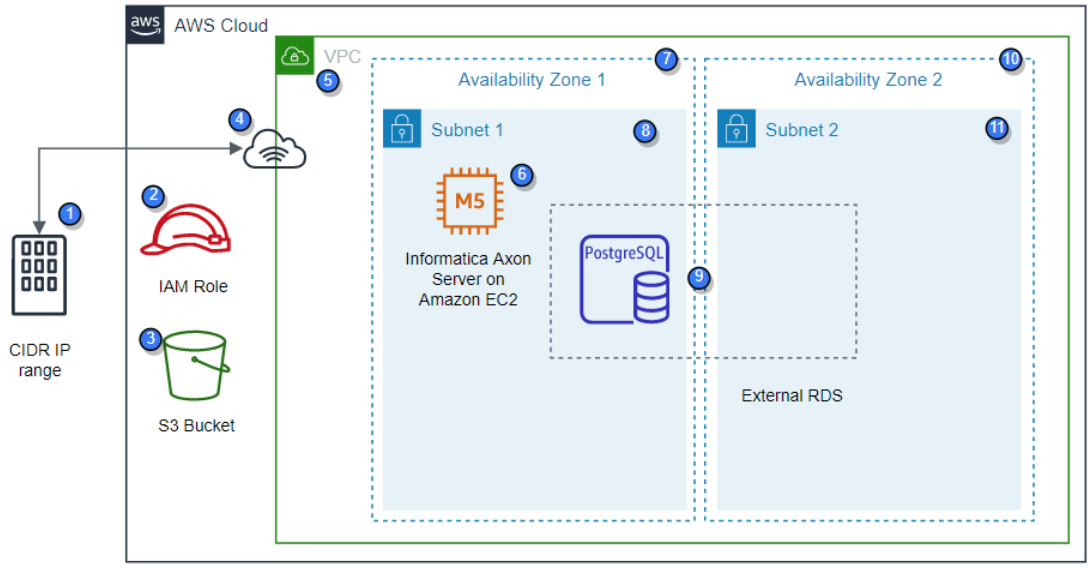
This deployment requires a license for Axon Data Governance. To sign up for a license, contact Informatica.

The following table lists the instance types that are used for the deployment:

Virtual Machine	Instance Type
Informatica Axon Server	M5.2xlarge/M4.2xlarge

Architecture

The following image shows the architecture of Axon Data Governance on the AWS U.S. Intelligence Community Marketplace:



1. Your company or home network CIDR IP range.
2. IAM roles.
3. S3 storage that acts as a location for the license key.
4. A gateway to enter AWS VPC resources. It can be a NAT gateway, Internet gateway, or a virtual private gateway.
5. A virtual private cloud (VPC) configured in the Availability Zone to contain the Axon Data Governance deployment.
6. The Informatica Axon Data Governance server.
7. Availability Zone 1. This deployment provisions one subnet.
8. Subnet 1 to contain specific elements, such as Amazon EC2 and PostgreSQL databases, of the deployment. Creates one subnet.
9. External PostgreSQL RDS databases.
10. Availability Zone 2. This deployment provisions one subnet.
11. Subnet 2 to contain specific elements, such as external PostgreSQL RDS, of the deployment. Creates one subnet.

Components in a Deployment

The deployment process creates the following components:

Component	Number of Components Created
Security groups	The following security groups: - One security group for the Axon Data Governance instance. - If you choose to create an external RDS, one security group for the external RDS instance.
Amazon EC2 instances	Informatica Axon Data Governance with an EC2 instance and 100 GB of storage space. The instance created is of type m5.2xlarge or m4.2xlarge, depending on availability of the instance size in the selected region.
IAM role and instance profile	1

Informatica Axon Data Governance

The Axon application consists of the core Axon component, services and microservices, and databases. These components exchange information through REST APIs.

Axon uses services and microservices to provide specific functionality that enable you to implement data governance. These services are independent, modular services that work together to enhance Axon capabilities.

Axon Data Governance includes the following services:

Service	Description
Authorization	Provides access to Axon objects based on the segments that are assigned to users.
Bulk Upload	Uploads a large number of Axon objects and saves them to the PostgreSQL database.
Change Request	Provides a controlled and audited process to make changes to Axon objects. Axon uses the Camunda service to manage and run workflows within a change request.
Data Marketplace	Provides access to Axon Data Marketplace. In Axon Data Marketplace, you can search and access data collections that Axon Data Governance users publish.
Document Upload	Attaches and maintains reference documents to an Axon object.
Notification	Sends notifications about changes to user roles, status of workflows and change requests, and status of bulk upload tasks.
Unison	Performs the Unison search. When you run a Unison search, Axon searches for the objects and the linked objects in all the available facets
Value List	Creates or updates a large number of values for data set attributes and saves the values to the PostgreSQL database.

Informatica Axon Quick Look

Informatica Axon Quick Look is a client program that you can quickly access from the desktop. Use Axon Quick Look to search and look up glossary terms that are defined in Axon Data Governance. You can also select keywords from applications, such as Microsoft Word, Microsoft Excel, and Google Chrome, and use the hotkeys to perform a lookup in the Axon Quick Look application.

Prerequisites

Before you launch the automated deployment on the AWS U.S. Intelligence Community Marketplace, verify the prerequisites and make the choices described in this section.

Network Prerequisites

Before you deploy Axon Data Governance, verify that you have completed the following network prerequisites for the Informatica application services. When you deploy to an existing VPC, the automated deployment provisions Axon Data Governance in your existing AWS infrastructure.

VPC prerequisites

Complete the following prerequisites when you choose to deploy into a VPC:

- Ensure that an IAM role and an IAM instance profile with the same name is present with ListBucket, GetObjects and ListObjects permissions. When you deploy into a VPC, ensure that you use the same bucket and folders which will be part of the IAM role's policy resources.
- Ensure that the VPC has DNS resolution enabled.
- Attach the subnet to a route table with a local route to the VPC CIDR.
- If the selected subnets are private, ensure that they can communicate to AWS services via VPC endpoints, NAT gateway or any other connectivity options that are available.
- Verify that each subnet in the VPC is associated with a route table.
- Verify that each route listed in the route table is defined as one of the following gateway types:
  - Internet gateway
  - NAT gateway
  - Virtual private gateway
- Verify that the subnet is associated with one of the following gateways:
  - Internet gateway
  - NAT gateway
  - Virtual private gateway
- Verify that the service subnets you choose are attached to a route table under the selected VPC. Also ensure that S3 and CloudFormation endpoints are setup for the selected VPC and subnets.
- Verify that your VPC has a minimum of two subnets.

Subnets require an Internet gateway, an NAT gateway, or a virtual private gateway in their route tables, to allow the instances to download packages and software without exposing them to the Internet. Record the domain name configured in the DHCP options as explained in the Amazon VPC documentation. You provide the domain name in the VPC settings when you configure the deployment.

Account and Security Prerequisites

Complete the prerequisites in this section to verify that you have the required AWS account and private keys.

AWS Account and Region

If you do not have an AWS account, create an account at aws.amazon.com.

Use the region selector in the navigation bar to choose the AWS region where you want to deploy Axon Data Governance on AWS.

Note: The region you choose determines the availability of certain sizes of EC2 instances.

If necessary, request a service limit increase for the EC2 instance type that you want to use for the Informatica domain. You might need to do this if you already have an existing deployment that uses this instance type and you think you might exceed the default limit with this reference deployment.

Private Keys

When you log in to an EC2 instance, use a password file for authentication. The file is called a private key file and has a file name extension of .pem.

If you do not have an existing .pem key to use, see the AWS documentation to create a key pair in your preferred region. A key pair is a set of security credentials that consists of a public key and a private key.

Note: Your administrator might ask you to use an existing key pair.

When you create a key pair, you save the .pem file to your desktop system. Simultaneously, AWS saves the key pair to your account. Make a note of the key pair that you want to use for the deployment, so that you can provide the key pair name during network configuration.

Account User Policies

Ensure that the AWS user account that you use has administrator privileges or the following policies assigned to the AWS user account:

- arn:aws:iam::aws:policy/AWSCloudFormationFullAccess
- arn:aws:iam::aws:policy/AmazonEC2FullAccess
- arn:aws:iam::aws:policy/IAMFullAccess
- arn:aws:iam::aws:policy/AmazonS3FullAccess
- arn:aws:iam::aws:policy/AmazonRDSFullAccess
- arn:aws:iam::aws:policy/AmazonVPCFullAccess

Storage Prerequisites

Create an Amazon S3 bucket for the Axon Data Governance license.

Amazon S3 stores objects that consist of one or more files and accompanying file metadata in buckets. For information about Amazon S3, see the AWS documentation.

To use the automated solution on the cloud marketplace, you must create an S3 bucket to store the license. You can store the license on the bucket level or in a folder within the S3 bucket.

Note the location of the license key and supply it when you configure the solution in the **License Key S3 Bucket URI** parameter.

Deployment Options

The automated deployment of Axon Data Governance on the AWS U.S. Intelligence Community Marketplace uses the AWS CloudFormation console to guide your choices and launch the solution deployment.

The deployment lets you configure additional settings such as CIDR blocks and Axon Data Governance settings, as discussed later in this guide.

Use the AWS CloudFormation console to specify the stack name and to enter parameter values to the template that you use to deploy the application. A stack is a logical representation of an AWS deployment.

Specify the identifiers for the VPC, for the subnet that contains the Amazon EC2 instances where the deployment process deploys the application components, and for the subnet where the deployment process creates the Model repository. The deployment process provisions the application components and creates an Axon server instance.

Availability zones and subnets

The automated solution deploys the components and services for Axon Data Governance into an Amazon VPC.

In the first subnet, the deployment process provisions Axon Data Governance on an Amazon EC2 instance within the VPC.

For more information about Axon Data Governance, see the *Axon Data Governance Data Governance Administrator Guide* and *Axon Data Governance Data Governance User Guide*.

Deployment Steps

You use a two-step process to deploy the Axon Data Governance 7.2.3 solution on the AWS U.S. Intelligence Community Marketplace.

Step 1: Get a license for Axon Data Governance

Perform the following steps to get a license:

1. This deployment requires a license for Axon Data Governance. To sign up for a license, contact Informatica.
2. In your AWS account, create an S3 bucket.
3. Optionally, create a directory under the S3 bucket.
4. Place the Axon Data Governance license key file for the software in the S3 bucket or directory.

Step 2: Launch the deployment

Each deployment takes approximately one to two hours to complete.

Perform the following steps to launch the deployment:

1. Use the AWS account to log in to the AWS Marketplace for U.S. Intelligence Community portal.
2. Type **Axon Data Governance** in the search box and press **Enter** on your keyboard.
3. Click **Axon Data Governance**.

   The **Product Overview** page appears.
4. Click **Continue to Subscribe**.

   The **Subscribe to this software** page appears.
5. Click **Continue to Configuration**.

   The **Configure this software** page appears.
6. To deploy Axon Data Governance on an existing VPC, select **Axon Data Governance** from the **Delivery Method** field.
7. Choose **7.2.3** from the **Software Version** list.
8. Click **Continue to Launch**.

   The **Launch this software** page appears.
9. Select **Launch CloudFormation** from the **Choose Action** list and click **Launch**. The **Create stack** page appears.
10. On the **Specify template** section, keep the default setting for the template URL, and then choose **Next**.

    The default option is `Amazon S3 URL`.
11. On the **Specify stack details** page, change the stack name if needed. Review the parameters for the template. Provide values for the parameters that require input.

12. Under **Capabilities**, select the check box to acknowledge that the template will create IAM resources.
13. To deploy the stack, click **Create**.

    Amazon AWS begins to create the stack and displays the CloudFormation dashboard. When the status is CREATE_COMPLETE, the deployment is complete. You can use the URL displayed on the **Output tab** to view the stack properties.
14. For all other parameters, review the default settings and customize them as necessary. When you finish reviewing and customizing the parameters, choose **Next.**

Parameters for deployment

1. On the **Create Stack** page, enter a name for the stack. The stack name can include letters (A-Z and a-z), numbers (0-9), and dashes (-).
2. Enter the parameters required to deploy the application.
   The following table lists the parameters in the **Network Configuration** section:

Parameter	Description
VPC	Required. ID of your existing VPC where you want to deploy Axon Data Governance.
Subnet 1	Required. Select the subnet ID where you want to deploy Axon Data Governance. Make sure that the subnet is included in the VPC and associated with an Internet Gateway or a NAT Gateway. Ensure that service subnet ID1 and service subnet are not the same.
Subnet 2	Required. CIDR block for the private subnet 2. Additional subnet is required for the external RDS deployment.
CIDR IP Range	Required. The CIDR IP range that is permitted to access Axon Data Governance. To reduce the potential of inbound attacks from unknown IP addresses, you can use a constrained CIDR range. For example, to specify the range of 10.20.30.40 to 10.20.30.47, enter 10.20.30.40/29.

The following table lists the configuration parameters for the **External PostgreSQL RDS** database:

Parameter	Description
Deploy PostgreSQL RDS	Select `Yes` to deploy an external PostgreSQL RDS database. Default is `No`.
DB admin username	The Admin user name for the database instance. Default is `adminuser`.
DB admin password	Required. The Admin password for the database instance.

The following table lists the parameters in the **Amazon EC2 Configuration** section:

Parameter	Description
Key Pair Name	Required. Name of an existing Amazon EC2 key pair to enable SSH and Remote Desktop Protocol (RDP) access to Axon Data Governance.
Instance Profile	Required. Name of an existing instance profile with the same name as the IAM role to be attached to the EC2 instance.

The following table lists the parameters in the **Axon Data Governance** Configuration section:

Parameter label (name)	Description
License Key S3 Bucket URI	Required. The S3 bucket URI for the Axon Data Governance license key in the following format: `s3://<bucketname>/<directory>/<license key>`

3. Click **Next** after you configure the parameters in the Axon Data Governance Configuration section.
4. On the **Configure stack options** page, you can specify key-value pairs for resources in your stack and set advanced options. When you are done, choose **Next**.
5. On the **Review** page, review and confirm the template settings. Under Capabilities, select the check box to acknowledge that the template will create IAM resources.
6. Choose **Create** to deploy the stack.
7. Monitor the status of the stack. When the status is `CREATE_COMPLETE`, the deployment is complete.

For information about logging in to Axon Data Governance, see the *Axon Data Governance Data Governance User Guide*.

Before You Begin

Before you start using Axon Data Governance, Informatica recommends that you modify your login password for security purposes. The following table contains the login credentials for the default administrator profile:

Default Administrator Username	Default Password
admin@informatica.com	Changeme@123

The password must be at least eight characters long and contain at least one lowercase character, one uppercase character, one numeric digit, and one special character. For more information about how you can change the default password, see the *Change Default Administrator Password* topic in the *Axon Data Governance Data Governance Administrator Guide*.

Monitoring Instance Provision and Informatica Axon Creation

You can use cloud platform dashboards, logs, or other artifacts to see whether cluster creation succeeded and how to locate and identify Axon Data Governance on the cloud platform.

Deployed AWS Resources

You can monitor the progress of the solution deployment in the Create Stack section of the AWS portal. Each row of the display shows stack components and creation status. The stack is complete when the status reads CREATE_COMPLETE.

When you deploy the solution to an existing VPC, the console displays the following stack components:

Component	Description
StackName/AxonStack	The main stack that is created by default.
StackName-SecurityGroupStack	Creates security groups and network interfaces used by EC2 instances. Created by default.
StackName-RDSStack	Create a PostgreSQL database instance for an external PostgreSQL module for Axon. Created if the **Deploy PostgreSQL RDS** parameter is set to **Yes**.
StackName-AxonServerStack	Creates an Informatica gateway node on which Axon Data Governance runs on an EC2 instance. Created by default.

Output Tab Properties

Click the **Output** tab to view the output parameters after the stack is successfully launched.

The following table lists the outputs:

Output	Description
AxonUser	User name of the Axon user.
AxonPassword	Password for the Axon user.
AxonQuickLookInstaller	The download location for the Axon Quick Look client installer.
InformaticaAxonURL	Private link with the port for the Axon application hosted on a private server.
InformaticaAxonLogs	Path to the Axon installation log.
InstanceSetupLogs	Path to the domain EC2 host setup log.
RDSEndpoint	The PostgreSQL RDS instance host name created by deployment. Available if the **Deploy PostgreSQL RDS** parameter is set to **Yes**.

For more information about the Axon Data Governance administrator tasks, see the *Axon Data Governance Data Governance Administrator Guide*.

Deployed Informatica Resources

The list of deployed resources includes the **InformaticaAxonURL**. To access the Informatica administrator, click the value of the **InformaticaAxonURL** property.

You can access the Axon application from the deployment output section.

If you are within same AWS virtual network, you can directly access the Axon application.

Solution Deployment Log Files

You can review the following logs on the Axon Data Governance to see the solution deployment details:

- Service Initialization log file: `/var/log/cfn-init.log`
- Command Initialization log file: `/var/log/cfn-init-cmd.log`
- CloudFormation log file: `/var/log/cfn-wire.log`
- Cloud log file: `/var/log/cloud-init.log`
- Command execution log file: `/opt/Infa_OnceClick_Solution.log`
- Node preparation log file: `/opt/Infa_node_configure.log`
- Axon Configuration log file: `/opt/Informatica/AxonServer/Configure.log`
- Axon Installation log file: `/opt/Informatica/AxonServer/Informatica_Axon_7.2.3.log`

Note: The command execution log records the creation of Informatica Axon Data Governance. If any of the tasks fail to complete, you can look at the detailed section for the task to troubleshoot.

Troubleshooting

Review the AWS log files to troubleshoot issues with your Axon Data Governance deployment.

Stack Failure

If an EC2 instance where Informatica services run is running but the stack fails, or if the Administrator tool fails to launch, review the Node preparation log file, Installation log file, Command Initialization log file, Service Initialization log file and Cloud log file to troubleshoot the issue.

For more information about the aforementioned log files, see "Solution Deployment Log Files" on page 12.

Frequently Asked Questions on Deployment Error Messages

Review the following frequently asked questions to understand the error messages that you encounter when you deploy Axon on AWS Marketplace:

I encountered a CREATE_FAILED error when I launched the deployment. What should I do?

If AWS CloudFormation fails to create the stack, we recommend that you relaunch the template with **Rollback on failure** set to **No**. (This setting is under **Advanced** in the AWS CloudFormation console, **Options** page.) With this setting, the stack's state will be retained, and the instance will be left running, so you can troubleshoot the issue. (You will want to look at the log files in `/var/log/cfn-init-cmd.log`)

Note: When you set **Rollback on failure** to **No**, you will continue to incur AWS charges for this stack. Please make sure to delete the stack when you have finished troubleshooting.

I encountered errors while installing Axon Data Governance. Where can I find more information?

View the `/opt/Infa_OneClick_Solution.log` log file to see more information about the errors that you encountered.

I encountered a size limitation error when I deployed the AWS CloudFormation templates.

We recommend that you launch the deployment templates from the location we provided or from another S3 bucket. If you deploy the templates from a local copy on your computer or from a non-S3 location, you might encounter template size limitations when you create the stack. For more information about AWS CloudFormation limits, see the AWS documentation.

What should I do before I relaunch the template with Rollback on failure?

Ensure that you disable rollback when you launch the stack in case of a stack failure. If Axon Data Governance server stack is created, check the following logs to identify the issues:

- `/opt/Infa_node_configure.log`
- `/opt/Infa_OneClick_Solution.log`
- `/var/log/cfn-init.log`

Author